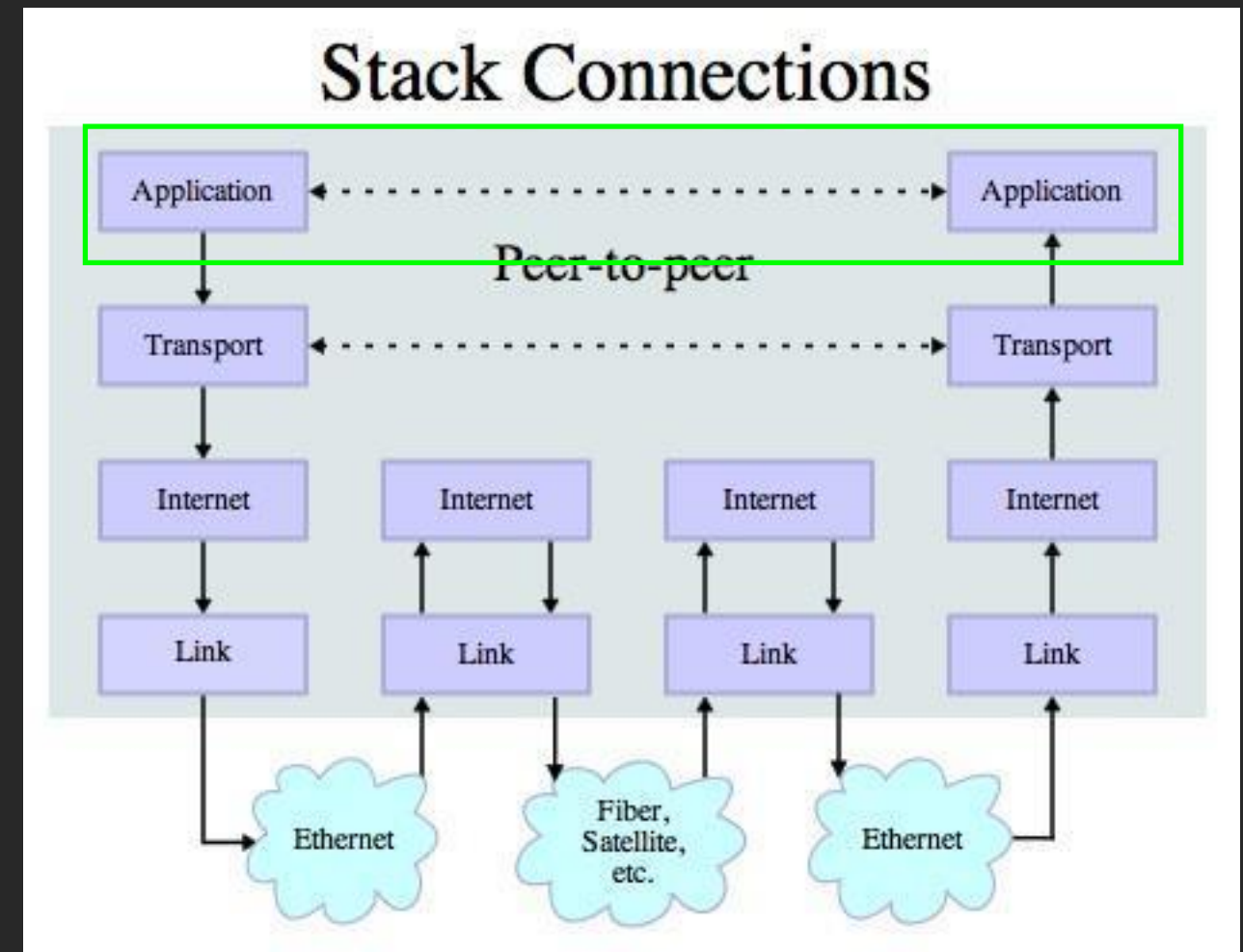


Application Protocol

- Since TCP (and Python) gives us a reliable **socket**, what do we want to do with the **socket**? What problem do we want to solve?
- Application Protocols
 - Mail
 - World Wide Web



Source: http://en.wikipedia.org/wiki/Internet_Protocol_Suite

HTTP - Hypertext Transfer Protocol

- The dominant Application Layer Protocol on the Internet
- Invented for the Web - to Retrieve HTML, Images, Documents, etc
- Extended to be data in addition to documents - RSS, Web Services, etc..Basic Concept - Make a Connection - Request a document - Retrieve the Document - Close the Connection

<http://en.wikipedia.org/wiki/Http>

HTTP

The **H**yper**T**ext **T**ransfer **P**rotocol is the set of rules to allow browsers to retrieve web documents from servers over the Internet

What is a Protocol?

- A set of rules that all parties follow so we can predict each other's behavior
- And not bump into each other
 - On two-way roads in USA, drive on the right-hand side of the road
 - On two-way roads in the UK, drive on the left-hand side of the road



`http://www.dr-chuck.com/page1.htm`

`protocol`

`host`

`document`

<http://www.youtube.com/watch?v=x2GylLq59rl>

1:17 - 2:19



Getting Data From The Server

Each the user clicks on an anchor tag with an href= value to switch to a new page, the browser makes a connection to the web server and issues a “GET” request - to GET the content of the page at the specified URL

The server returns the HTML document to the Browser which formats and displays the document to the user.

Web Server

80



Browser

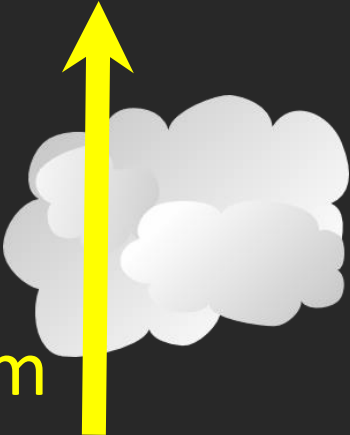




Click

Browser

Request



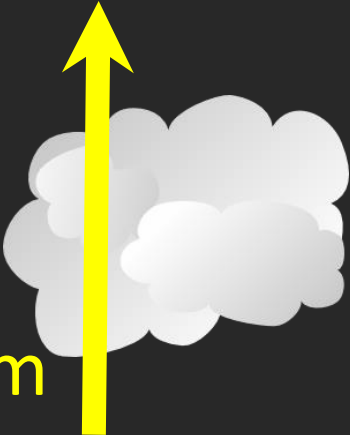
GET <http://www.dr-chuck.com/page2.htm>

Browser



Click

Request



GET <http://www.dr-chuck.com/page2.htm>

Browser

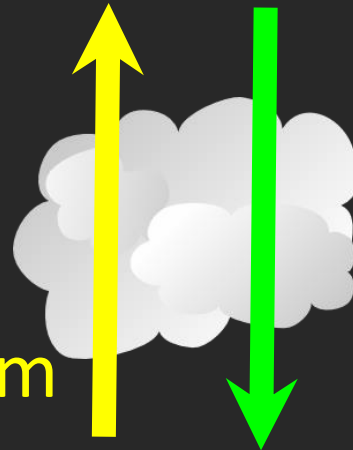


Click

Request

Web Server

80



GET http://www.dr-chuck.com/page2.htm

Response

```
<h1>The Second  
Page</h1><p>If you like, you  
can switch back to the <a  
href="page1.htm">First  
Page</a>.</p>
```

Browser



Click

Request



Response

```
<h1>The Second
Page</h1><p>If you like, you
can switch back to the <a
href="page1.htm">First
Page</a>.</p>
```

GET http://www.dr-chuck.com/page2.htm

Browser



Click

Parse/
Render



Internet Standards

- The standards for all of the Internet protocols (inner workings) are developed by an organization
- Internet Engineering Task Force (IETF)
- www.ietf.org
- Standards are called “RFCs” - “Request for Comments”

INTERNET PROTOCOL

DARPA INTERNET PROGRAM

PROTOCOL SPECIFICATION

September 1981

The internet protocol treats each internet datagram as an independent entity unrelated to any other internet datagram. There are no connections or logical circuits (virtual or otherwise).

The internet protocol uses four key mechanisms in providing its service: Type of Service, Time to Live, Options, and Header Checksum.

Source: <http://tools.ietf.org/html/rfc791>

<http://www.w3.org/Protocols/rfc2616/rfc2616.txt>

Network Working Group
Request for Comments: 2616
Obsoletes: 2068
Category: Standards Track

R. Fielding
UC Irvine
J. Gettys
Compaq/W3C
J. Mogul
Compaq
H. Frystyk
W3C/MIT
L. Masinter
Xerox
P. Leach
Microsoft
T. Berners-Lee
W3C/MIT
June 1999

Hypertext Transfer Protocol -- HTTP/1.1

Status of this Memo

This document specifies an Internet standards track protocol for the Internet community, and requests discussion and suggestions for improvements. Please refer to the current edition of the "Internet Official Protocol Standards" (STD 1) for the standardization state and status of this protocol. Distribution of this memo is unlimited.

Copyright Notice

Copyright (C) The Internet Society (1999). All Rights Reserved.

Abstract

The Hypertext Transfer Protocol (HTTP) is an application-level protocol for distributed, collaborative, hypermedia information

5 Request

A request message from a client to a server includes, within the first line of that message, the method to be applied to the resource, the identifier of the resource, and the protocol version in use.

```
Request          = Request-Line           ; Section 5.1
                  *(( general-header       ; Section 4.5
                    | request-header       ; Section 5.3
                    | entity-header ) CRLF) ; Section 7.1
                  CRLF
                  [ message-body ]        ; Section 4.3
```

5.1 Request-Line

The Request-Line begins with a method token, followed by the Request-URI and the protocol version, and ending with CRLF. The elements are separated by SP characters. No CR or LF is allowed except in the final CRLF sequence.

```
Request-Line     = Method SP Request-URI SP HTTP-Version CRLF
```

Making an HTTP request

Connect to the server like `www.dr-chuck.com`"

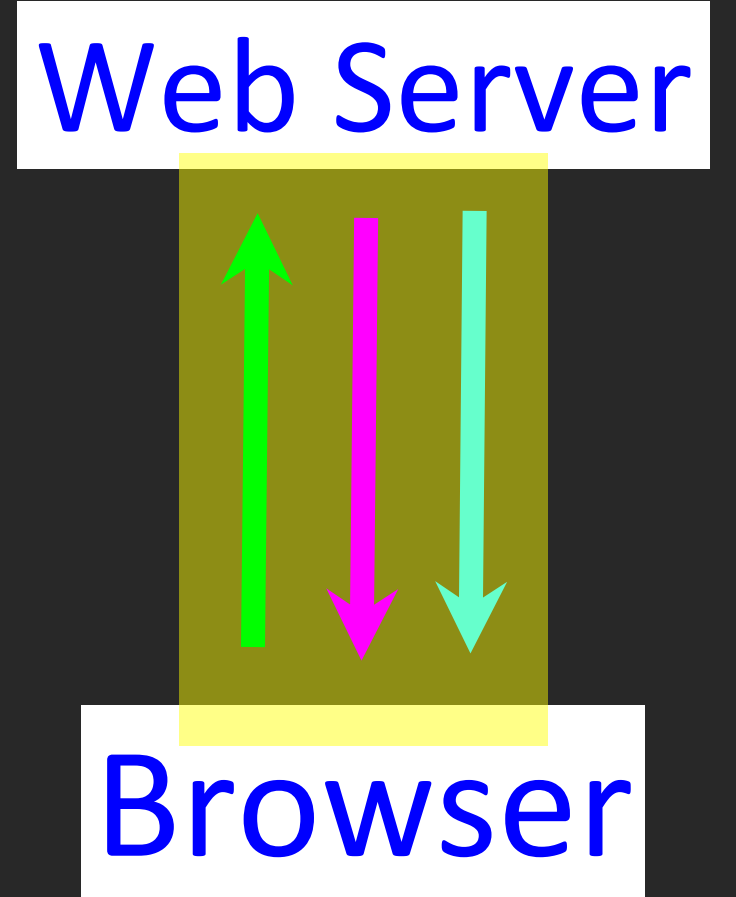
Request a document (or the default document)

- *GET http://www.dr-chuck.com/page1.htm HTTP/1.0*
- *GET http://www.mlive.com/ann-arbor/ HTTP/1.0*
- *GET http://www.facebook.com HTTP/1.0*

```
$ telnet data.pr4e.org 80
Trying 74.208.28.177...
Connected to data.pr4e.org.
Escape character is '^]'.
GET http://data.pr4e.org/page1.htm HTTP/1.0

HTTP/1.1 200 OK
Date: Tue, 30 Jan 2024 15:30:13 GMT
Server: Apache/2.4.18 (Ubuntu)
Last-Modified: Mon, 15 May 2017 11:11:47 GMT
Content-Length: 128
Content-Type: text/html

<h1>The First Page</h1>
<p>If you like, you can switch to
the <a href="http://data.pr4e.org/page2.htm">Second
Page</a>.</p>
Connection closed by foreign host.
```



Web Server

Browser

Accurate Hacking in the Movies

Matrix Reloaded
Bourne Ultimatum
Die Hard 4

...

<http://nmap.org/movies.html>



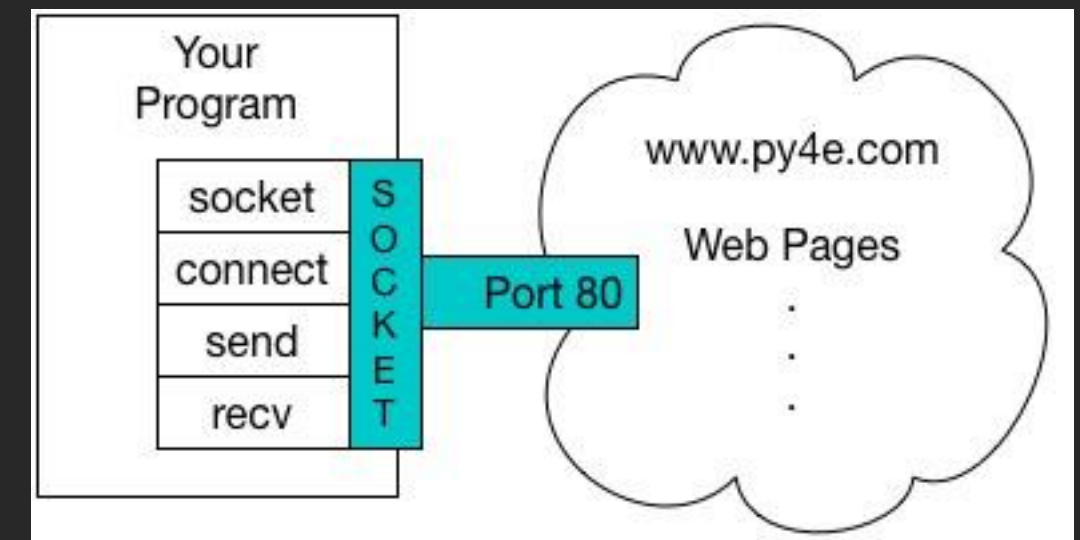
```
80/tcp    open      http
81/tcp    open      hosts2-ns
10.0.0.1  [mobile]
11 # nmap -u -ss -O 10.2.2.2
11
13 Starting nmap V. 2.54BETA25
13 Insufficient responses for TCP sequencing (3). OS detection i
13 accurate
14 Interesting ports on 10.2.2.2:
44 (The 1539 ports scanned but not shown below are in state: clo
51 Port      State      Service
51 22/tcp    open      ssh
58
68 No exact OS matches for host
68
24 Nmap run completed -- 1 IP address (1 host up) scanned
50 # sshnuke 10.2.2.2 -rootpw="Z10H0101"
Connecting to 10.2.2.2:ssh ... successful.
Re Attempting to exploit SSHv1 CRC32 ... successful.
IP Resetting root password to "Z10H0101".
System open: Access level (9)
Hn # ssh 10.2.2.2 -l root
root@10.2.2.2's password:
RTI CONTROL
ACCESS GRANTED
```

An HTTP Request in Python

```
import socket

mysock = socket.socket(socket.AF_INET, socket.SOCK_STREAM)
mysock.connect(('data.pr4e.org', 80))
cmd = 'GET http://data.pr4e.org/romeo.txt HTTP/1.0\n\n'.encode()
mysock.send(cmd)

while True:
    data = mysock.recv(512)
    if (len(data) < 1):
        break
    print(data.decode())
mysock.close()
```



```
HTTP/1.1 200 OK
Date: Sun, 14 Mar 2010 23:52:41 GMT
Server: Apache
Last-Modified: Tue, 29 Dec 2009 01:31:22 GMT
ETag: "143c1b33-a7-4b395bea"
Accept-Ranges: bytes
Content-Length: 167
Connection: close
Content-Type: text/plain
```

```
But soft what light through yonder window breaks
It is the east and Juliet is the sun
Arise fair sun and kill the envious moon
Who is already sick and pale with grief
```

HTTP Header

```
while True:
    data = mysock.recv(512)
    if ( len(data) < 1 ) :
        break
    print(data.decode())
```

HTTP Body



About Characters and Strings...



Acknowledgements / Contributions



These slides are Copyright 2010- Charles R. Severance (www.dr-chuck.com) of the University of Michigan School of Information and open.umich.edu and made available under a Creative Commons Attribution 4.0 License. Please maintain this last slide in all copies of the document to comply with the attribution requirements of the license. If you make a change, feel free to add your name and organization to the list of contributors on this page as you republish the materials.

Initial Development: Charles Severance, University of Michigan School of Information

... Insert new Contributors here